Worksheet 6-8 & 6-7 A Basic Indefinite Integrals #1

AP Calculus Name_____

Directions: Integrate the following expressions. Simplify all answers.

1. $\int \left(x^3 - 2 + \frac{7}{x}\right) dx =$

2. $\int (x^2 + 3x - 5e^x)\, dx =$

3. $\int (x^6 - 5x^3 + x)\, dx =$

4. $\int (4x^5 - 6)\, dx =$

5. $\int \left(\frac{2}{3}x^3 - \frac{1}{2}x^2 + 6x\right) dx =$

6. $\int \frac{4}{\sqrt[3]{t}} dt = =$

7. $\int \left[\frac{2\sqrt[5]{x}}{5} - \frac{7}{2\sqrt[4]{x}}\right] dx =$

8. $\int \left[\frac{1}{2x} - 5^x\right] dx$

9. $\int \left(4 - x^2 + 5\sqrt[3]{x}\right) dx =$

10. $\int \left(2\sqrt[3]{t} - 5\sqrt{t}\right) dt =$

11. $\int \left(\frac{2}{\sqrt[5]{z}} + \frac{1}{3\sqrt[3]{z}}\right) dz =$

12. $\int -4 \sec x \tan x\, dx =$

13. $\int 5 \sin x\, dx =$

14. $\int -\frac{2}{3} \csc^2 y\, dy =$

15. $\int -\frac{4 \cos m}{7} dm =$

16. $\int (2 \cos x + x^2) dx =$

17. $\int \left(\sin x + 3\sqrt{x}\right) dx =$

18. $\int \frac{\sin x}{\cos^2 x} dx =$

19. $\int \sec t\,(\sec t + \tan t)\, dt$

20. $\int \left(\frac{1}{x^2} + \frac{2}{x^3}\right) dx =$

21. $\int \left(\frac{3}{2x^4} - \frac{5}{x^2}\right) dx =$

22. $\int \left(\frac{1}{x^5} - \frac{2}{3x^3}\right) dx =$

---

23. $\int \left(5 + \frac{2}{3}x^2 + \frac{3}{4}x^3\right) dx =$

24. $\int \left(u^6 - 2u^5 - u^3 + \frac{2}{7}\right) du =$

25. $\int (u+4)(2u+1)\ du =$

26. $\int \sqrt{t}(t^2 + 3t + 2)\, dt =$

27. $\int \frac{1 + \sqrt{x} + x}{x}\ dx =$

28. $\int \left(\frac{1+r}{r}\right)^2 dr =$